SIMATS ENGINEERING

ASSESSMENT TOOL 3: Mock Interview

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1506

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 25**
Question Count: 10

Code of Conduct

I M. Pranay (192525382)_certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay ____________

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong	Reasonable reasoning	Basic reasoning	Weak reasoning

		architecture-level reasoning			
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

ANSWERS

1. Difference between HDFS and Traditional File Systems

- **HDFS** is a distributed file system designed for very large datasets.
- It divides files into blocks and stores them across multiple DataNodes.
- It provides automatic **data replication and fault tolerance**.
- Traditional file systems generally store files on a single computer or storage system.
- HDFS is designed for **high-throughput processing** rather than low-latency access.
- HDFS uses a **NameNode** for metadata and **DataNodes** for actual data storage.

In short: HDFS provides distributed, scalable and fault-tolerant storage for big data.

2. Why is Replication Important?

Replication means maintaining multiple copies of the same data block.

- Protects data against node or disk failure.
- Improves data availability.
- Provides fault tolerance.
- Allows processing to continue even when one DataNode fails.
- Can improve read performance because data can be read from another replica.

Example: With a replication factor of 3, three copies of a block are maintained on different DataNodes.

3. How does MapReduce Divide Work?

MapReduce divides processing into two major phases:

Map Phase

- Input data is divided into input splits.
- Mapper processes each split independently.
- It generates intermediate **key-value pairs**.

Shuffle and Sort

- Intermediate results are grouped according to their keys.
- Related values are sent to the appropriate reducer.

Reduce Phase

- Reducers process grouped key-value pairs.
- They generate the final output.

Example: For word counting, the mapper produces (word, 1) and the reducer adds all values for each word.

4. Role of YARN
YARN stands for **Yet Another Resource Negotiator**.
Its main responsibilities are:

- Manages cluster resources.
- Allocates CPU and memory to applications.
- Schedules jobs.
- Monitors running applications.
- Supports multiple processing frameworks.
- Improves resource utilization.

Main components include:

- **ResourceManager**
- **NodeManager**
- **ApplicationMaster**
- **Containers**

Thus, YARN separates **resource management from data processing**.

5. NAS vs SAN

NAS	**SAN**
Network Attached Storage	Storage Area Network
File-level storage	Block-level storage
Accessed using network file protocols	Accessed as storage blocks
Easier to manage	More complex
Suitable for file sharing	Suitable for high-performance applications
Generally lower cost	Generally higher cost

HDFS differs from both because it distributes data across multiple commodity servers.

6. Purpose of ETL
ETL means **Extract, Transform and Load**.

- **Extract:** Collect data from different sources.
- **Transform:** Clean, validate and convert data into a required format.
- **Load:** Store processed data in the target system.

ETL helps integrate data from databases, applications, files and other sources before analytics.
Example: Customer data from multiple systems can be extracted, duplicate records removed, and the cleaned data loaded into a big data platform.

7. How Apache NiFi Helps Data Integration

Apache NiFi is a data-flow and integration platform.
It helps to:

- Collect data from multiple sources.
- Route data between systems.
- Transform data.
- Monitor data flows.
- Track data provenance.
- Support real-time data movement.
- Connect databases, APIs, files and big data systems.

A typical flow is:
Source → NiFi → Transformation → Validation → HDFS/Data Lake → Analytics

8. Key Failure Points in Hadoop

Important failure points include:

- **DataNode failure**
- **Disk failure**
- **NameNode failure**
- Network failure
- ResourceManager failure
- Application/task failure

Handling methods:

- Use HDFS replication.
- Use NameNode high availability.
- Replace failed disks/nodes.
- Use automatic task re-execution.
- Use ResourceManager recovery.
- Monitor cluster health continuously.

This ensures high availability and fault tolerance.

9. Secure Data Ingestion Pipeline

A secure pipeline can be designed as:
Data Sources → Secure Ingestion → Authentication → Encryption → Validation → HDFS → Access Control → Analytics

- Use Apache NiFi or Kafka for ingestion.
- Authenticate data producers and users.
- Encrypt data during transmission.
- Encrypt sensitive data at rest.
- Validate incoming data.
- Apply role-based access control.
- Maintain audit logs.

- Monitor unauthorized access.
- Apply data retention policies.

This protects data throughout its lifecycle.

10. Explain Hadoop to a Non-Technical Manager

I would explain it simply:

"Hadoop is like a team of computers working together. Instead of asking one powerful computer to store and process a huge amount of data, Hadoop divides the data and workload among many computers."

- **HDFS** stores large amounts of data across multiple computers.
- **MapReduce** divides a large processing task into smaller tasks.
- **YARN** manages computer resources and decides which tasks should run where.
- More computers can be added when data volume increases.